DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**UMBILICAL CATHETERIZATION**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-NICU-024 (v2)**		
Last Version Approval Date	**28/1/2022**	Number of Pages	**6**

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APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
19 / 3 / 2024	7 / 4 / 2024	TRIENNIAL	18 / 3 / 2027

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1.0 PURPOSE

1.1. To assist the physician with insertion of an umbilical catheter and monitor the newborn during the procedure. The principal indication for umbilical vein catheterization is to gain vascular access during emergency resuscitation or fluid administration.

1.2. To obtain reliable arterial access for blood sampling.

2.0 DEFINITION/ABBREVIATION

2.1. **Umbilical vein catheterization** is an insertion of a catheter through the umbilical vein that may be a life-saving procedure in neonates who require vascular access and resuscitation. The best location is beyond the ductus venous in the central venous system (inferior vena cava or at the junction of right atrium).

2.2. **Umbilical artery catheterization** is a procedure wherein a catheter is inserted via on of the umbilical artery to the premeasured desired position (between T6-T9).

2.3. **Indication of umbilical artery:**

2.3.1. Arterial blood sampling

2.3.2. Arterial blood pressure monitoring

2.3.3. Administration of fluids

2.3.4. Exchange transfusions (removal of blood)

2.4. **Indication of Umbilical Vein**:

2.4.1. Intravenous access for the administration of fluid, drugs, or hypertonic glucose.

2.4.2. To gain vascular access during emergency resuscitation

2.4.3. Monitoring central venous pressure

2.4.4. Venous blood sampling

2.4.5. To perform exchange transfusion

3.0 RESPONSIBILITY

2.1. **Staff Nurse** – prepare equipments, assisting physician in safe insertion of umbilical catheter under strict aseptic technique.

2.2. **Physician** – assisted by the nurse under strict aseptic technique, must insert umbilical catheterization.

4.0 POLICY

4.1. Umbilical catheterization is a procedure in which a radiopaque catheter is passed through an umbilical artery, or through the umbilical vein. After proper placement of the umbilical line,

intravenous fluids and medication may be administered to critically ill neonates. Umbilical catheterization can be an effective method of administering therapeutic fluids and agents or of obtaining diagnostic blood samples from a high-risk newborn, but great care is required in inserting and monitoring the tube.

4.2. Umbilical catheterization must be inserted by the physician, assisted by the nurse under strict aseptic technique.

4.3. The umbilical catheter should be made flexible, non-toxic radiopaque material that will not kink when advanced through a vessel and will not collapse during blood withdrawal.

4.4. Before initiating the procedure, a radiant warmer should be obtained, and the infant should be connected to a cardiac monitor.

4.5. The location and proper placement of the umbilical catheter must be confirmed by X-ray before infusing medications or IV fluids. X-ray is recommended once or twice weekly to confirm placement of lines.

4.6. The umbilical catheter should be filled with heparinized saline before insertion.

4.7. Umbilical catheters must be properly labeled, identifying UAC catheters from the UVC catheters.

4.8. Packed red blood cells (PRBCs) should not be given or infused through Umbilical arterial catheter. Umbilical Arterial lines can compromise circulation to lower extremities. If toes or feet become dusky, line should be assessed for clot. If line flushes and draws easily, but duskiness persists or worsens, line should be immediately removed and the limb watched closely for vascular compromise. Signs of emboli to the lower extremities are an indication to remove the catheter.

4.9. Antibiotic peaks and troughs should not be drawn from the line through which they were administered.

4.10. If the umbilical line is displaced, pressure should be applied quickly to the cord with a sterile 4 x 4 gauze, and notify the physician immediately.

4.11. All connections to the umbilical line should be checked every 30 to 60-minute. Condition of the cord should be observed every 2 to 3 hours for signs of infection such as redness, edema, or drainage at the catheter insertion site.

4.12. Abdominal positioning should be used cautiously with UAC/UVC's since accidental slipping, kinking, or removal of the catheter may occur without being immediately apparent.

5.0 PROCEDURE (S)

5.1. Perform hand hygiene.

5.2. Open the umbilical catheter tray and add 10 ml syringe, stopcock and suture using aseptic technique to facilitate a well-organized procedure ensuring sterility.

5.3. Place infant in supine position. Wrap a diaper around both legs to restrain the infant to stabilize the patient for the procedure.

5.4. Ensure that the newborn is connected to cardiopulmonary monitor.

5.5. Document vital signs and oxygen saturation to provide a baseline assessment.

5.6. Both the nurse and physician should wear gloves, gown, caps and masks maintaining aseptic technique to facilitate sterility of the procedure.

5.7. Clean the port of heparin solution with alcohol swabs to ensure sterility.

5.8. Hold the heparin solution while the scrub nurse draws solution into 10 ml syringe.

5.9. Physician will flush the stopcock and catheter.

5.10. The physician or scrub nurse will determine the length of catheter insertion depth.

5.10.1. UAC = 3x weight in Kg + 9 cm

5.10.2. UVC = ½ of UAC + 1 cm

5.10.3. Do not forget to add the length of the stump to length.

5.11. Clean the umbilical cord with chlorhexidine. In the extremely preterm newborn the chlorhexidine must be removed from the skin with saline. Do not remove the chlorhexidine from the umbilical stump.

5.12. Drape the newborn with sterile towels.

5.13. Assist the physician to perform the procedure:

5.13.1. Tie a piece of umbilical tape around the base of the umbilical cord, tight enough to minimize blood loss but loose enough so that the catheter can be passed easily through the vessel. Cut off the excess umbilical cord with scissors or scalpel leaving 1 cm. stump. A scalpel usually makes a cleaner cut, so that the vessels are more easily seen. There are 2 umbilical arteries and one umbilical vein.

5.13.2. Using a curve haemostat grasp the end of the umbilicus to hold upright and steady. Use the forceps to open and dilate umbilical artery then insert the umbilical catheters. Aspirate to confirm blood return.

5.14. Attach prepared IV tubing to the stopcock, turn stopcock off to sterile 3 ml syringe.

5.15. Obtain X- ray as ordered to verify correct placement, before infusing IV fluids.

5.16. The physician will suture the catheters to the umbilical stump, and tape the umbilical catheters using bridge/tape to form H to secure the umbilical catheters.

5.17. The umbilical stump with the catheter in place is left open. No special dressing is needed.

5.17.1. Infection

5.17.2. Vascular accident

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5.17.3. Haemorrhage

5.18. Connect the IV fluids and administer medication as ordered to the umbilical line.

5.19. Monitor the newborn throughout the procedure for changes in vital signs such as desaturations, bradycardia and apnea.

5.20. Observe the lower extremities for vasospasm. Monitor the buttocks and/or toes for belching, cyanosis or color changes, especially when the arterial line is flushed. Notify the doctor to provide prompt intervention, because these suggest compromised blood supply.

5.21. If vasospasm is observed, inform the physician and warmed the opposite leg. If the vasospasm does not resolve within 30 mints. to 1 hour, informed the physician again to assess immediately as the line needs to be removed.

5.22. Document the time procedure started and completed position of the catheter, the doctor who inserted the lines, catheter size and newborn's tolerance of the procedure and if there is any complication.

6.0 EQUIPMENTS

6.1. Umbilical catheter tray (needle holder, irish forceps without teeth, scissor)
6.2. Umbilical catheters (Fr.-2.5, 3.5, 5) for arterial line
6.3. Double lumen catheter (Fr. 4) for venous line
6.4. 3 ml and 10 ml syringes
6.5. Heparin flush solution
6.6. Neonatal infusion pump & tubing
6.7. Sterile gown and gloves
6.8. Blade size 15 / scalpel
6.9. 4-0 silk suture (physician preference)
6.10. Tape to secure catheter
6.11. Antiseptic solution
6.12. Umbilical tie
6.13. Sterile saline
6.14. Sterile towels
6.15. Caps and masks
6.16. Tape measure
6.17. Stopcock
6.18. 18-gauge needle
6.19. Gauze pads

7.0 ATTACHMENT/FORMS

7.1. Nurses Progress Notes (GDOH-NUR-NPN-203)
7.2. Nurse Observation Form (Form No. 43)

8.0 REFERENCES

8.1. Guidelines for Neonatal Care, (NICU) MOH Saudi Arabia

8.2. Carole Kenner, Judy Wright Lott, (2007). Comprehensive Neonatal Nursing. (3th Ed.) Saunders.

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8.3. John P.Coherty, Eric C. Eichenwald, Ann R. Stark. (2008). Manual of Neonatal Care. (6th Ed.). Wolters Kluwer/ Lippincott Williams & Wilkins

8.4. Mhairi G. MacDonald, Jayashree Ramasethu, (2007) Atlas of Procedure in Neonatology (3rd Ed.). Lippincott Williams & Wilkins.

9.0 REVISION CLAUSE

9.1. Modification of policy lay-out, coding and numbering

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KINGDOM OF SAUDI ARABIA

وزارة الصحة
Ministry of Health

Hospital: _____ مستشفى:

Region: _____ المنطقة/المحافظة:

Dept./Unit: _____ القسم/الوحدة:

MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

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SN

Ministry of Health
General Directorate of Health Affairs
Al-Baha

وزارة الصحة
المديرية العامة للشئون الصحية بمنطقة الباحة

HOSPITAL :

Health Care No .: ☐☐☐☐☐☐☐☐☐☐☐☐

Name : .....

Age : ☐☐ Yrs./Month Sex : ☐M ☐F

Nationality : .....

Consultant In-Charge : .....

Dept : ..... Unit : .....

NURSES OBSERVATION FORM — نمـوذج مـلاحظـات التمـريـض

Nurse / Date	Time	Temp.	Puls Rate	Resp Rate	B.P.	Pupils		Level of Consciousness Muscle Tone &Reflex	Cough Reflex	Response to Stimuli		Perception Pulses			Skin Color	Rema-rks	Sign.
						R	L			Verbal	Painful	ORAL	Audio Aural	visual			

SPECIFIC NURSING CARE COMMENTS :

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